

# Physical Examination

All pressure ulcer patients should undergo a full physical examination to identify systemic diseases contributing to wound development, such as anemia, chronic cardiac or respiratory disease, and neurologic disorders.

## Ulcer Evaluation

Staging is an assessment system that classifies pressure ulcers based on the anatomic depth of soft-tissue damage. Pressure ulcers often progress from lower to higher stages, and any small pressure ulcer should be considered the possible "tip of an iceberg." Palpation is important to assist visual evaluation, and debridement is often necessary to determine the true extent of ulceration. Ulcers that initially appear superficial may be classified as stage III or IV after debridement reveals deeper tissue involvement. The epidermis may show hypertrophy at the ulcer margin with varying degrees of pigmentation.

Several staging systems are used to classify pressure ulcers. The most commonly used include the National Pressure Ulcer Advisory Panel (NPUAP), Shea, and Yarkony and Kirk systems. These systems differ in criteria and categorization but aim to standardize assessment and guide treatment.

## Related Physical Findings

Tenderness, erythema, edema, warmth of surrounding skin, exudate, and foul odor are signs of local infection. Fever, along with declining mental or physical status, should raise suspicion for bacteremia or osteomyelitis.

Spasticity resulting from inflammation or infection may lead to muscle contractures and joint deformities, limiting mobility and complicating patient

positioning. Weakness, anemia, and dehydration may also be present due to significant fluid and protein loss from chronic, draining wounds.

# Clinical Findings

## History

A thorough history and physical examination for pressure ulcer patients should include the use of a validated risk assessment scale. The Braden, Waterlow, and Norton scales are the most widely used tools for predicting the risk of developing pressure ulcers. These scales have been extensively tested for reliability and validity and are recommended by the Agency for Healthcare Research and Quality. Key risk factors assessed include:

- Mobility
- Activity
- Moisture exposure
- Nutritional status
- Friction and shear
- Sensory perception

# Laboratory Tests

The diagnosis of pressure ulcers is primarily clinical. However, laboratory abnormalities such as anemia, leukocytosis, hypoproteinemia, or low serum iron levels may be present and reflect underlying systemic compromise. It is important to rule out other conditions that can cause skin ulcers, including calciphylaxis, vasculitis, and connective tissue diseases.

# Biopsy

Histologic analysis can be useful when clinical evaluation is insufficient to determine ulcer depth or to exclude other ulcerative conditions. In chronic pressure ulcers, there is a risk of malignant transformation, particularly to squamous cell carcinoma, known as Marjolin's ulcer. Histopathologic findings in pressure ulcers are detailed in relevant clinical tables.

# Imaging Studies

Imaging is most valuable for identifying complications of deep pressure ulcers.

- **Plain radiographs** can detect ectopic bone, gas in the ulcer cavity (suggesting infection), and sclerotic or destructive changes in underlying bone.
- Ectopic bone formation may contribute to ulcer development by altering posture and increasing localized pressure.
- When the extent of deep tissue involvement is unclear, **sinography** may be performed to delineate sinus tracts or fistulous connections.